

AVDASI2 PROJECT WORKSHOP

Down-Selection & Risk Management
Week 3 – Monday 6th October 2025
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Convergent tools - Learning objectives

Unit level learning objective(s)

2. Carry out the design, build and test of a functioning major UAV assembly as part of a team, **using applicable interdisciplinary concepts and methods;**



Session level learning objectives

1. **Recall** the benefits of using a structured down-selection methodology
2. **Utilise** a *numerical* and *non-numerical pairwise comparison tool* to **rank** or **weight** solution criteria
3. **Utilise** a *controlled convergence* tool to **select** an optimum solution
4. **Utilise** a *multi-criteria decision analysis (MCDA) method* to **select** an optimum solution

Video

This
live
session



Convergent tools – Recap

Tool 1 = Pairwise comparison (Simple)

Function		A	B	C	D	E	F		Count	Rank (1 = n)
Strength	A		AB	A	D	E	A	Count of A	2.5	4
Appearance	B			C	E	E	F	Count of B	0.5	6
Speed	C				C	C	F	Count of C	3	2
Carbon footprint	D					D	F	Count of D	2	5
Design resource	E						E	Count of E	4	1
Durability	F							Count of F	3	3

Use for ranking criteria

Evaluate criteria

Tool 2 = Pairwise comparison (Numerical)

[illegible]

Use for assigning a weighting factor to your design criteria

Tool 3 = Controlled convergence

Evaluate designs

Tool 4 = Multicriteria decision analysis

[illegible]

Comparing designs
against chosen criteria
(=, better or worse)

MCDA matrix										
Criteria	Weightings	Designs →	design a		design b		design c		design d	
			score	weighted score	score	weighted score	score	weighted score	score	weighted score
Initial Cost	0.100	1	4	0.4	3	12.0	1	3.0	1	1.0
Durability	0.150	2	1	0.2	2	2.0	3	6.0	2	6.0
Weight	0.050	3	3	0.2	1	3.0	5	5.0	5	25.0
Handling	0.100	4	2	0.2	5	10.0	0	0.0	5	0.0
blank1	0.100	5	4	0.4	0	0.0	0	0.0	0	0.0
blank2	0.100	6	5	0.5	0	0.0	0	0.0	0	0.0
blank3	0.400	7	6	2.4	0	0.0	0	0.0	0	0.0
	Total score for design →		4.2		27.0		14.0		32.0	

Comparing designs, using weighted criteria and a scoring system

Combining tools

Pairwise comparison matrix										
			=Much More Important	=More Important	The =Same	=Less Important	=Much Less Important			
Choose a scale (symmetric) →			9	3	1	0.333333	0.1111111111			
Criteria		Initial Cos	Durability	Weight	Handling	Disposal	0	0	Total	%
	X	1	2	3	4	5	6	7		
Initial Cost	1	X	1	1	3	3			8.00	24
Durability	2	1	X	1	1	3			6.00	18
Weight	3	1	1	X	9	3			14.00	41
Handling	4	0.333333	1	0.111111	X	0.333333			1.78	5
Disposal	5	0.333333	0.333333	0.333333	3	X			4.00	12
0	6						X		0.00	
0	7							X	0.00	
		2.67	3.33	2.44	16.00	9.33	0.00	0.00	33.78	100
		7.9	9.9	7.2	47.4	27.6	0.0	0.0	100	33.78

Weightings from this
Pairwise comparison ...

...can be used as the
weightings in the MCDA

MCDA matrix										
Criteria	Weightings	Designs →	design a		design b		design c		design d	
	↓ Must add to 100% ↓		score	weighted score	score	weighted score	score	weighted score	score	weighted score
Initial Cost	0.240	1	4	1.0	3	0.7	1	0.2	1	0.2
Durability	0.180	2	1	0.2	2	0.4	3	0.5	2	0.4
Weight	0.410	3	3	1.2	1	0.4	5	2.1	5	2.1
Handling	0.050	4	2	0.1	5	0.3	0	0.0	5	0.3
Disposal	0.120	5		0.0		0.0		0.0		0.0
0		6		0.0		0.0		0.0		0.0
0		7		0.0		0.0		0.0		0.0
	1									
Total score for design →			2.5		1.7		2.8		2.9	

Highest score suggests optimum design

Combining tools

Warning – scales can cause excessive sensitivity.

Pairwise comparison matrix									
			=Much More Important	=More Important	The =Same	=Less Important	=Much Less Important		
Choose a scale (symmetric) →			9	3	1	0.333333	0.1111111111		
Criteria		Initial Cost	Durability	Weight	Handling	Disposal	0	0	
	X	1	2	3	4	5	6	7	Total %
Initial Cost	1	X	1	1	3	3			8.00 24
Durability	2	1	X	1	1	3			6.00 18
Weight	3	1	1	X	9	3			14.00 41
Handling	4	0.333333	1	0.111111	X	0.333333			1.78 5
Disposal	5	0.333333	0.333333	0.333333	3	X			4.00 12
0	6						X		0.00 0
0	7							X	0.00 0
		2.67	3.33	2.44	16.00	9.33	0.00	0.00	33.78 100
		7.9	9.9	7.2	47.4	27.6	0.0	0.0	100 33.78

MCD matrix

Criteria	Weightings	Designs →	design a		design b		design c		design d	
	↓ Must add to 100% ↓		score	weighted score	score	weighted score	score	weighted score	score	weighted score
Initial Cost	0.240	1	4	1.0	3	0.7	1	0.2	1	0.2
Durability	0.180	2	1	0.2	2	0.4	3	0.5	2	0.4
Weight	0.410	3	3	1.2	1	0.4	5	2.1	5	2.1
Handling	0.050	4	2	0.1	5	0.3	0	0.0	5	0.3
Disposal	0.120	5		0.0		0.0		0.0		0.0
0		6		0.0		0.0		0.0		0.0
0		7		0.0		0.0		0.0		0.0
	1									
Total score for design →			2.5		1.7		2.8		2.9	

Developing tools further

MCDA matrix

Criteria	Weightings	Designs →	design a		design b		design c		design d	
↓ Must add to 100% ↓			score	weighted score	score	weighted score	score	weighted score	score	weighted score
Initial Cost	0.240	1	4	1.0	3	0.7	1	0.2	1	0.2
Durability	0.180	2	1	0.2	2	0.4	3	0.5	2	0.4
Weight	0.410	3	3	1.2	1	0.4	5	2.1	5	2.1
Handling	0.050	4	2	0.1	5	0.3	0	0.0	5	0.3
Disposal	0.120	5		0.0		0.0		0.0		0.0
0		6		0.0		0.0		0.0		0.0
0		7		0.0		0.0		0.0		0.0
1										
Total score for design →			2.5		1.7		2.8		2.9	

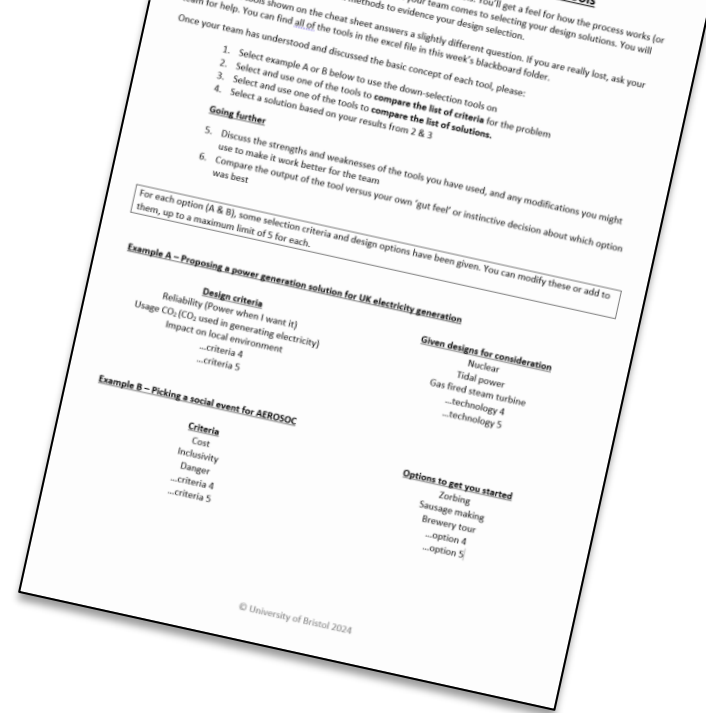
Could this be an average based on team votes?

Is more democratic necessarily better?

Should each voice carry equal weight?

4. Exercise – Convergence

- Read the worksheet instructions
- Pick which question (A or B) to carry out the down-selection on
- Decide/complete your list of criteria
- Decide/complete your list of potential solutions
- Select which tools are appropriate for getting you an answer & then use the tools
- Discuss the result of the tool output vs your initial gut feel
- Consider any limitations of the methods used
- Consider any opportunities to improve the method



Learning objectives

Unit level learning objective(s)

4. discuss key health and safety responsibilities for engineers; and using recognised risk management tools create risk assessments to analyse a variety of project risks;

Session level learning objectives



1. **Discuss** the Health and Safety responsibilities of engineers operating in the UK, **referring** to the key legal requirements & implications

-
- ✓ 2. **Create** a risk assessment using a recognised process whilst **identifying** frequency scores, consequence scores; and appropriate risk mitigations for operational, technical and project risks
 - ✓ 3. **Generate** a scoring system for the classification of frequency and consequence
 - 4. **Classify** risks using a criticality matrix

Risk Assessment



Risk Assessment

Moral/Ethical

Health and Safety

Inclusion, Ethics,
Sustainability



(these also have legal
requirements here)

Economic

Schedule (Delays)

Financial (\$)

Reputational

Damage to property
/ equipment

Legal (Law)

Commercial
/Corporate

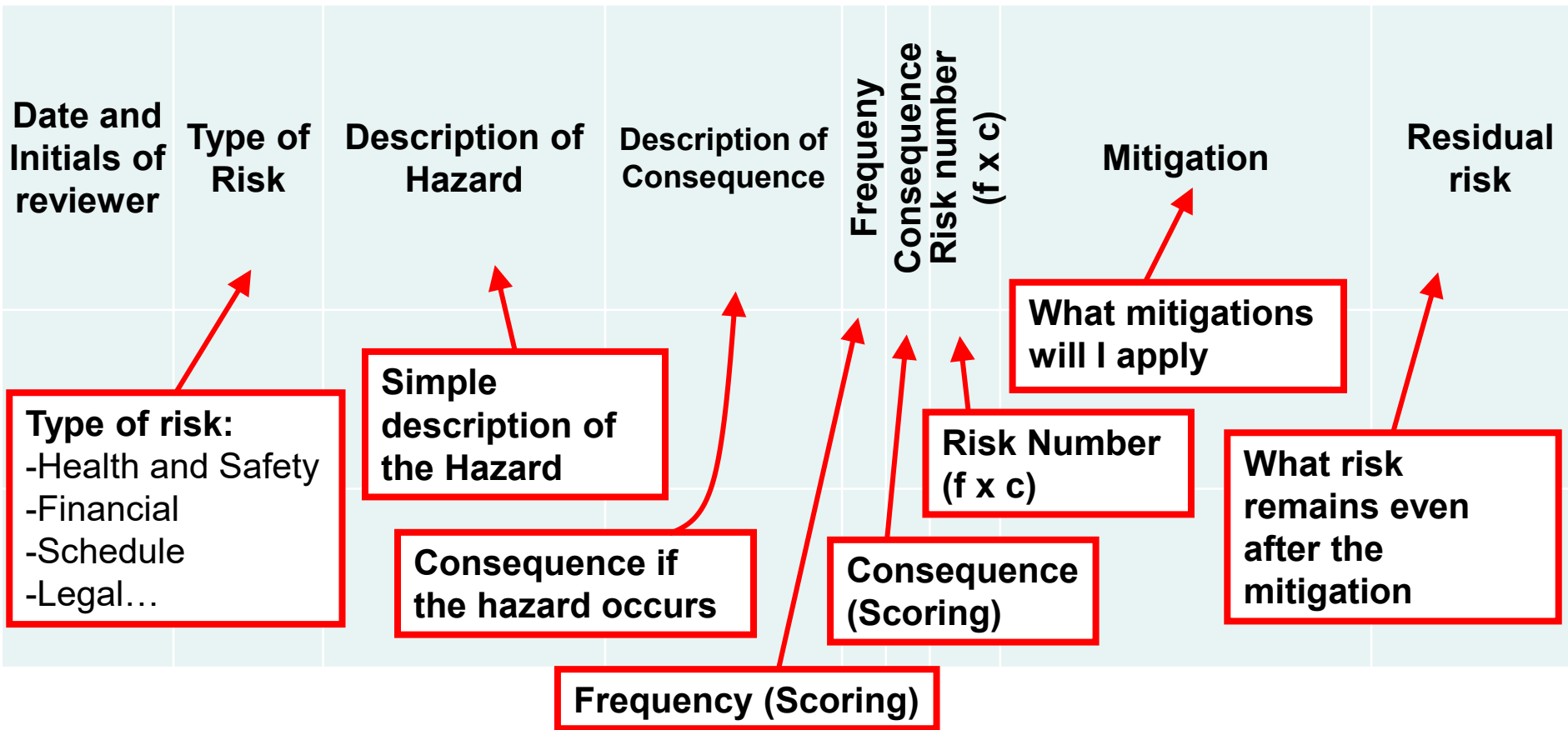
Criminal

Civil

Intellectual
property

Employment

Completing a Risk Assessment



Risk Assessment – examples

Hazard =
something that
has potential to
cause harm

Hazard...**Total**
failure of
mounting system
during overnight
stay on
'Portaledge'.

Hazard...**Partial**
failure of
mounting system
during overnight
stay on
'Portaledge'.



Risk Assessment – examples

Frequency =
how often/likely
is this likely to
happen



Hazard = Server outage
due to power interruption

1 = < 1 in 10,000 hrs
2 = 1 in 10,000 hrs
3 = 1 in 1,000 hrs
4 = 1 in 100 hrs
5 = 1 in 10 hrs

Frequency = ?

Select
a scale
of 1-5
or 1-10



Hazard = Landing gear stuck up due to
total failure of the hydraulic hose

1 = 1 in 100,000 components
2 = 1 in 10,000 "
3 = 1 in 1,000 "
4 = 1 in 100 "
5 = 1 in 10 "

Frequency = ?

Risk Assessment – examples

Consequence

= if this does happen, what might be the consequences

Select a scale
of 1-5 or 1-10



Hazard = Test rig failure.
Delay to product release.

Financial impact

- 1 = Negligible (<\$10k)
- 2 = Minor (>\$10k <\$100k)
- 3 = Major (>\$100k <\$1m)
- 4 = Severe (>\$1m <\$10m)
- 5 = Extreme >\$10m

Consequence = ?



31st October 2021, Salisbury.
Image from RAIB

Hazard = Train fails to stop due to wheel slide causing collision.

- 1 = (Negligible) Non-reportable injury
- 2 = (Minor) Minor Injury
- 3 = (Major) Major injury or multiple minor injuries
- 4 = (Critical) Single fatality or multiple major injuries
- 5 = (Catastrophic) Multiple fatalities

Risk Assessment – examples

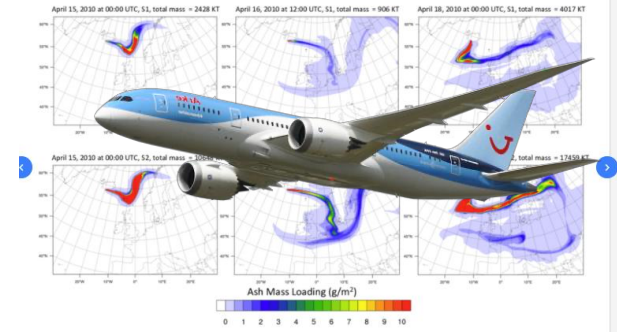


Risk Number
= frequency x
consequence

Hazard = Ingestion of volcanic ash into both engines



Eyjafjallajökull, Iceland 2010



Frequency (1) $\times$ Consequence (5) Risk number = 5

Hazard = Bird strike – 1 engine



Frequency (3) $\times$ Consequence (4) Risk number = 12

Risk assessment – Going further

Date and Initials of reviewer	Type of Risk	Description of Hazard	Description of Consequence	Frequency	Consequence	Risk number (f x c)	Mitigation	Residual risk

Consider
rescoring the
Frequency,
Consequence
and RN here is a
great idea.

Advanced Template - Post mitigation scoring

[illegible]

≈20 minutes - Develop a risk assessment

Example A) Use of **dry ice** to cool brakes on Formula 1 car during race preparation.



motorsport.com

Complete the risk assessment for 3-4 risks.

Include a range of risk types – not just safety.

Complete all a row before adding new risks

Example B) Drone inspection of wind turbine



Shutterstock.com

Use the blank paper copies.

Take 5 min to develop a scoring system before starting. It does not need to be perfect!

Tip (for recording existing and new mitigations)

Date and Initials of reviewer	Type of Risk	Description of Hazard	Description of Consequence	Frequency Consequence Risk number (f x c)	Mitigation	Residual risk
Best to split mitigations into already existing/deployed mitigations and additional mitigations (as a result of the RA). The score is based on existing mitigations.					What mitigations will I apply	
Example for climbing rope:		<i>Existing mitigation:</i> Use min 8kN rope to En 892 <i>Additional mitigation:</i> Deploy double rope system on traversing climbs.			Existing: Additional:	

Example Risk Assessment



ROYAL INTERNATIONAL AIR TATTOO 2024 EVENT RISK ASSESSMENT (B)

